HASSAN RAZA

AI Engineer | LLMs, RAG & Computer Vision | Python, FastAPI, Docker

Lahore, Pakistan | +92 324 7930605 | hassanraza.rashid5@gmail.com | github.com/HassanRaza5121 | linkedin.com/in/hassan-raza-436398222

PROFESSIONAL SUMMARY

AI Engineer with nearly four years of hands-on experience building and deploying production machine learning and generative AI systems, from data pipeline to deployment. Delivered RAG, fine-tuned LLM, and computer vision solutions with measurable results, including 98% recognition accuracy and a 75% cut in manual work. Skilled in Python, FastAPI, Docker, and CI/CD, and experienced in working directly with clients and mentoring engineers.

TECHNICAL SKILLS

Languages: Python, SQL

Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LLM Fine-Tuning, Prompt Engineering, NLP

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Computer Vision, Neural Networks, Feature Engineering

MLOps & Tools: Docker, FastAPI, REST APIs, CI/CD, Git, GitHub, Linux

PROFESSIONAL EXPERIENCE

AI / Machine Learning Engineer (Freelance), Remote *Jan 2023 – Present*

- **RAG system:** Built and deployed a Retrieval-Augmented Generation system over call-transcript data, enabling context-aware question answering and automated insight extraction.
- **LLM fine-tuning:** Fine-tuned the open-source TinyLlama language model for a mental health support chatbot, reaching 85% sentiment-detection accuracy across 500+ users.
- **Computer vision:** Implemented a face-recognition attendance system with 98% accuracy across varied conditions, cutting manual tracking time by 75% for a workforce of 200+.
- **Object detection:** Built an end-to-end object-detection pipeline from scratch, including dataset collection, labeling, feature engineering, and detection logic.
- **Risk prediction:** Developed a machine learning pipeline for financial risk prediction on large-scale transactional data, with automated feature engineering and model optimization.
- **Security ML:** Designed a phishing-detection system combining Neural Networks and SVMs, with features engineered from URL patterns and benchmarked across multiple threat scenarios.
- Applied MLOps practices (Docker, FastAPI, CI/CD) for training, evaluation, and deployment, and worked directly with clients to gather requirements and deliver solutions.

Teaching Assistant, Machine Learning, Information Technology University (ITU) *Jan 2024 – Present*

- Mentor students on neural networks, transformers, and optimization methods, and guide applied model development and experimentation.
- Run lab sessions and grade projects for ML/AI coursework.

Data Analyst Intern, Punjab Administration Authority *2024*

- Cleaned and analyzed large datasets and built dashboards to communicate trends and performance metrics to stakeholders.

EDUCATION

BS, Electrical Engineering, Information Technology University (ITU), Pakistan *2024*

Graduate Coursework, Computer Science, Information Technology University (ITU), Pakistan *2024 – 2025*

LEADERSHIP

- President, ITU Career Counselling Society: led workshops and events that reached 500+ students.